

TELCO CUSTOMER CHURN ANALYSIS REPORT (PROFESSIONAL VERSION)

1. Project Title

Telco Customer Churn Analysis using Exploratory Data Analysis (EDA)

2. Objective

The primary objective of this project is to analyze customer behavior data from a telecom company to identify key factors influencing customer churn.

Customer churn refers to customers who discontinue their subscription or stop using the service. Understanding churn patterns helps businesses improve customer retention strategies, reduce revenue loss, and enhance service quality.

3. Dataset Description

The dataset used in this analysis is the **Telco Customer Churn Dataset**, which contains **7,043 customer records** and **21 features** describing customer demographics, account information, and service usage.

Key Features:

Feature	Description
customerID	Unique identifier for each customer
gender	Customer gender (Male/Female)
SeniorCitizen	Whether customer is senior citizen
Partner	Whether customer has a partner
Dependents	Whether customer has dependents
tenure	Number of months customer stayed
PhoneService	Whether customer has phone service
InternetService	Type of internet service

Feature	Description
Contract	Contract type (Month-to-month, One year, Two year)
MonthlyCharges	Monthly billing amount
TotalCharges	Total amount charged
Churn	Whether customer left the company

4. 🧹 Data Cleaning & Preprocessing

Several preprocessing steps were performed to ensure data quality:

✓ Handling Missing Values

- The TotalCharges column contained missing or invalid entries.
- It was converted to numeric format.
- Missing values were replaced using the **median strategy**.

✓ Data Type Corrections

- Converted TotalCharges from object type to numeric.
- Ensured correct datatype consistency for analysis.

✓ Data Integrity Check

- Verified no duplicate records exist.
 - Confirmed dataset completeness after cleaning.
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5. 📊 Exploratory Data Analysis (EDA)

5.1 💎 Churn Distribution

- The dataset shows an **imbalanced distribution** of churned vs non-churned customers.
- Around **26% customers have churned**, while **74% remained active**.

👉 Insight: Imbalanced dataset indicates churn is a critical minority event.

5.2 ♦ Tenure Analysis

- Customers with **low tenure (0–12 months)** show highest churn rate.
- Long-term customers (> 24 months) are more stable.

👉 Insight: Early lifecycle customers are highly vulnerable.

5.3 ♦ Monthly Charges Analysis

- Customers with **higher monthly charges are more likely to churn.**
- Distribution shows churn increases with cost burden.

👉 Insight: Pricing strategy plays a major role in retention.

5.4 ♦ Contract Type Analysis

- **Month-to-month contracts** have the highest churn rate.
- **One-year and two-year contracts** significantly reduce churn.

👉 Insight: Long-term contracts improve customer retention.

5.5 ♦ Gender Analysis

- No significant difference in churn between male and female customers.

👉 Insight: Gender is not a strong predictor of churn.

5.6 ♦ Internet Service Analysis

- Customers using **Fiber Optic internet service show higher churn rates.**
- DSL users show relatively lower churn.

👉 Insight: Service quality or pricing of fiber optic may affect retention.

5.7 ♦ Correlation Analysis

- Positive correlation observed between:
 - MonthlyCharges ↔ Churn
- Negative correlation observed between:
 - Tenure ↔ Churn

📖 Insight:

- Higher tenure reduces churn probability
 - Higher charges increase churn probability
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6. Key Business Insights

From the analysis, the following key insights were derived:

● High Risk Factors for Churn:

- Month-to-month contracts
- High monthly charges
- Low tenure customers
- Fiber optic internet users

□ Retention Factors:

- Long-term contracts
 - High tenure customers
 - Lower monthly charges
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7. Business Recommendations

To reduce churn, the telecom company should:

✓ 1. Promote Long-Term Contracts

Encourage customers to switch from month-to-month to yearly plans.

✓ 2. Improve Pricing Strategy

Offer discounts or bundles for high-paying customers.

✓ 3. Customer Retention Programs

Focus on new customers (first 6–12 months) with offers and support.

✓ 4. Improve Fiber Optic Service Quality

Investigate issues causing higher churn in fiber users.

✓ 5. Personalized Offers

Use customer segmentation for targeted retention campaigns.

8. 📊 Visualizations Used

The following visualizations were used in this project:

- 📌 Churn Distribution Plot
- 📌 Tenure Histogram
- 📌 Monthly Charges Distribution
- 📌 Contract vs Churn Count Plot
- 📌 Internet Service vs Churn Plot
- 📌 Correlation Heatmap

All charts are saved in the /charts folder for reference.

9. 📁 Output Files Generated

File	Description
cleaned_telco_churn.csv	Cleaned dataset
analysis_report.md	Full analysis report
charts/*.png	Visualization outputs

10. 🚩 Conclusion

The analysis clearly shows that customer churn is strongly influenced by contract type, tenure, and monthly charges.

Customers with short tenure, high charges, and month-to-month contracts are the most likely to churn.

By focusing on retention strategies for these segments, telecom companies can significantly reduce churn and improve customer lifetime value.